



Boot loaders are notorious for causing grief while trying to boot into multiple operating systems. Using LILO you can gain total control of the booting process

BOOTS in Control

Booting into Linux is almost magical; the stream of messages popping past is a sign of a healthy operating system starting up, or it could be a chance of coaxing a sick install back to life. However, there is a little something that lies between the point where your BIOS flashes past and Linux boots up—the boot loader. In true Linux spirit, boot loaders are not just little Windows-style point-and-click menus, but come with some pretty sophisticated features. The two most common are GRUB and LILO. Here, we'll take a look under the hood of boot loaders—LILO in particular.

What's the boot?

You power up the PC, and the code that resides in the BIOS kicks in. It detects and initialises all devices, and then clears all the registers. Now the BIOS code picks up the data sitting in the MBR (Master Boot Record) of the first boot device—which you've set in the BIOS settings—and places it in the system memory. This is the boot loader, which then starts the process

of loading and transferring control to the operating system.

LILO is one of the oldest boot loaders used to boot into Linux. Modern versions of LILO, which ship with the recent distributions, come with a graphical boot-menu. In Red Hat Linux 8 and 9, you have a choice between the default loader GRUB, or LILO. The latter allows you to boot into a wide variety of operating systems, including most distributions of Linux and also Unix clones such as FreeBSD Unix and NetBSD Unix; you can also use LILO to boot into more esoteric operating systems such as GNU/HURD and Solaris.

Put on those booties

LILO can be installed while installing Linux in your system. The Red Hat Linux 9 installer, for example, will let you install the boot loader after partitioning your hard drive.

Before choosing a boot loader, make sure to set a separate */boot* partition when partitioning your hard drive. This is best when installing Linux to a partition larger

than 8 GB. If you have multiple hard drives, you should install the */boot* partition in the first (master) drive—*/dev/hda*—within the first 8 GB of the drive. While all modern BIOSes, and LILO itself support large drives, you should do this if you happen to get warning messages that indicate there might be a problem with booting into Linux.

In Red Hat Linux 9, LILO can be installed by changing the default boot loading options—in this case, GRUB. The installation program lets you choose between GRUB and LILO in the Boot Loader Configuration screen. To choose LILO, click on Change Boot Loader. Here you can choose to install LILO, GRUB, or no boot loader at all. Choose 'Use LILO as the boot loader'. Next, assuming that you are using multiple operating systems, you can select the default operating system to boot into. Select the 'Default' checkbox next to one of the operating systems listed. If you have other OSes installed, but they do not show up on this list, click on Add and add the operating system. You can also use the Edit and Delete options to